

Philosophy of Space and Time

Lecture 2: *Zeno to Euclid — Geometry and Space*

Chris Smeenk

Department of Philosophy, UCLA · April 2, 2026

Today's Plan

- 1 Zeno: Remaining Responses
 - Composition of the Continuum
 - The At-At Theory of Motion
 - Taking Stock
- 2 Euclid's Geometry
- 3 Non-Euclidean Geometries
- 4 Philosophical Consequences

Four Paradoxes

Paradoxes of Traversal

1. The Dichotomy
2. Achilles & the Tortoise

Can a journey composed of infinitely many sub-journeys ever be completed?

Paradoxes of Composition

3. The Arrow
4. Infinite Divisibility

How is time composed of instants? How is a line composed of points?

Where We Left Off

Last time we saw that **convergent series** (arguably) resolves Dichotomy and Achilles: an infinite sum of finite quantities *can* be finite.

Limit of a sequence: A number L such that the terms get arbitrarily close to L .

Sum of a series: Consider the **partial sums**:

$$S_n = a_1 + a_2 + \cdots + a_n.$$

If $S_1, S_2, S_3, \dots$ converges to a limit S , then *by definition* the infinite sum equals S .
(Note: not all series converge! Claim: series relevant to paradoxes do converge.)

Two paradoxes remain unanswered:

- **The Arrow:** if the arrow is motionless at every instant, when does it move?
- **Infinite Divisibility:** how can dimensionless points compose a line of finite length?

(And even for the Dichotomy, we asked whether the mathematics really settles the philosophical question.)

Recall: Infinite Divisibility Paradox

The Argument

ID-P1: A line segment is composed of infinitely many points.

ID-P2: The length of a segment equals the sum of its parts.

ID-P3: All line segments have a definite, finite length.

ID-P4: Each point has either (a) zero magnitude, or (b) some finite magnitude (say, ℓ).

Dilemma: both options in ID-P4 contradict ID-P3:

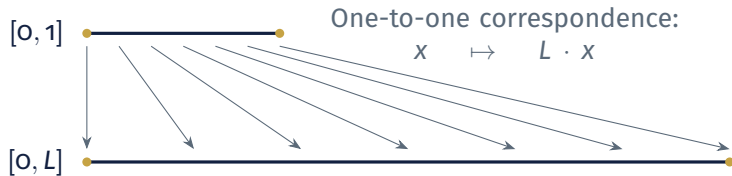
- If (a): $0 + 0 + 0 + \dots = 0$.
- If (b): $\ell + \ell + \ell + \dots = \infty$.

The Continuum Problem

The convergent-series answer does *not* resolve Infinite Divisibility:

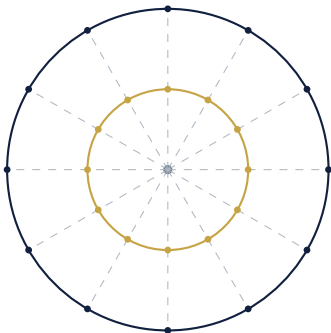
- There the sum is $\ell + \ell + \ell + \dots$ (equal terms) — this does *not converge*.
 - The puzzle is about the *composition* of the line, not about subdividing a journey.
- ? *How can a collection of dimensionless points constitute a line segment with definite, finite length?*

All Segments Have the Same “Number” of Points



Both segments contain the **same “number”** of points—regardless of length!

Concentric Circles



Each radial line pairs a point on the inner circle with a point on the outer.

Same “number” of points— but different circumferences!

Length $\neq$ counting points.

Countable vs. Uncountable Infinity

- **Countably infinite**: elements can be listed $a_1, a_2, a_3, \dots$ (one-to-one correspondence with $\mathbb{N}$). Examples: natural numbers; rationals.
- **Uncountably infinite**: Cantor proved (1874) that the real numbers *cannot* be listed this way.

The points on a line segment form an *uncountable* set.

Our definition of “infinite sum” only applies to **countable** sums — so the idea of summing up point-lengths to get the length of the line *does not apply*.

The Metric

The **length** (or *measure*) of a line segment is *not* determined by summing up the sizes of its constituent points.

- The set of points gives you the *topology* (which points are near which), but not the *metric* (how far apart they are).
- A **metric** is additional structure imposed *on top of* the point set.
- This is why segments with the “same number” of points can have different lengths.

Zeno assumes that length must be built from the lengths of points — the modern view rejects this.

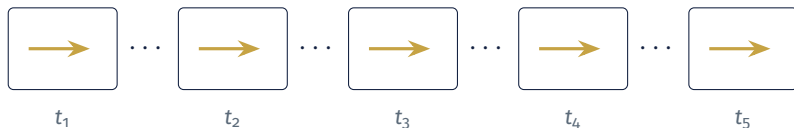
Recall: The Arrow Paradox

The Argument

A-P1: At each and every instant, the arrow is at rest.

A-P2: Any interval of time is composed of instants.

A-CNC: Therefore, during any interval, the arrow is not moving.



Response to the Arrow: The At-At Theory

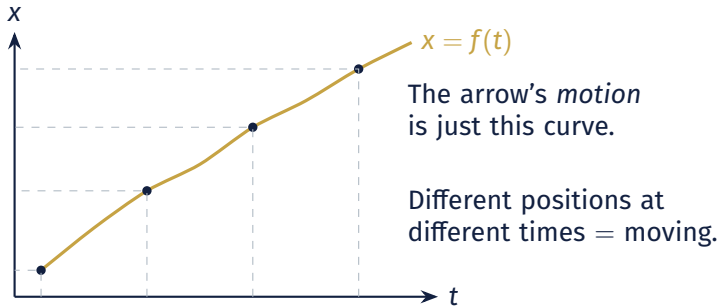
- Aristotle's response: deny A-P2, time is not composed of indivisible instants.
- A better move, first articulated by **Bertrand Russell**:

The At-At Theory

To be “in motion” is simply to be *at* the appropriate position *at* each moment of time. Motion is captured by a **function** $f: t \mapsto x(t)$.

The arrow is “moving” just in case f assigns it *different positions at different times*. No “process” at an individual instant is needed.

The At-At Theory — Visualized



Instantaneous Velocity

? *If the arrow doesn't “move” at a single instant, can we still say it has a velocity at that instant?*

Yes—the calculus provides a rigorous definition:

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} = f'(t)$$

- Instantaneous velocity = **derivative** of the position function.
- Defined via a limit involving *nearby* instants—not by what happens “during” a single instant.
- Even granting A-P1, an arrow can have a well-defined nonzero velocity at each instant.

Challenging A-P1

A deeper objection: **deny A-P1 outright.**

- A-P1 says the arrow is “at rest” at each instant.
- But perhaps rest *simply doesn't apply* to a single instant.
- Crucial difference between:
 - › “the arrow has zero velocity” (a substantive claim), and
 - › “the question of motion is inapplicable at a single instant.”
- The at-at theory says motion is defined *over intervals*; asking whether something moves “at” an instant is a **category error**.

Summary: Modern Responses to Zeno

Paradox	Core Problem	Modern Response
Dichotomy	Infinitely many sub-journeys	Infinite series can converge to a finite sum
Achilles	Faster never catches slower	Catching time = limit of convergent series
Arrow	Motionless at every instant	At-at theory: motion = function $t \mapsto x(t)$
Infinite Divisibility	Points sum to wrong length	Metric is extra structure on the point set

From Zeno to Geometry

The resolution of Infinite Divisibility introduced a key idea: the **metric** as additional structure that assigns lengths and distances.

- But what *is* this metric?
- These are the questions that **geometry** addresses.
- For over two thousand years, there seemed to be exactly one answer: *Euclid's geometry*.

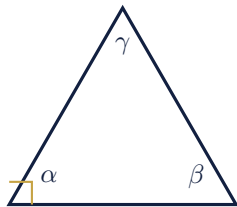
? *What is the geometry of physical space? And how could we know?*

Break

Euclid's Geometry

Euclid's *Elements* (c. 300 BCE)

- One of the most influential books ever written
- Model of the **axiomatic method**: start from a small number of self-evident truths, derive everything by logical deduction
- For over two thousand years, taken to describe the actual structure of physical space
- Paradigm of *certain* knowledge—known by pure reason



$$\alpha + \beta + \gamma = 180$$

(in Euclidean geometry)

Deductive Structure of the *Elements*

Euclid organizes geometry into a rigorous deductive system:

Definitions 23 geometrical terms: “A point is that which has no part.” “A line is breadthless length.” ...

Common Notions General principles of reasoning: “Things equal to the same thing are equal to one another.”

Postulates Specific geometric assumptions regarding possible constructions.

Propositions Problems (constructions) and theorems (proved assertions), built up step by step from the above.

If the axioms are *self-evidently true* and the derivations *logically valid*, the theorems are **certain knowledge**.

Status of Geometry

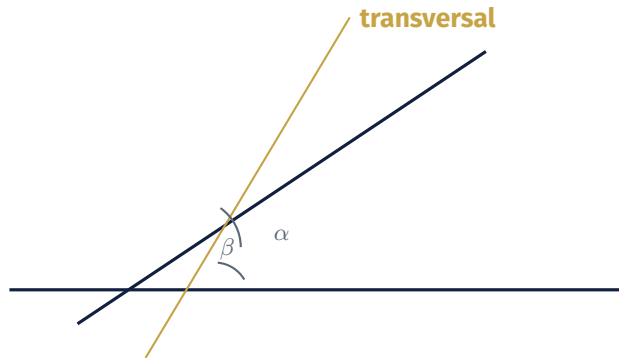
- An exemplary case of **demonstrative knowledge**:
 - › Theorems based on clear, undisputed definitions and postulates
 - › Transparent deductive structure showing how theorems follow
- This made geometry the gold standard for knowledge in Western philosophy—from Plato through Kant.
- ? *How is knowledge of this kind—synthetic rather than merely analytic—possible? What is the subject matter of geometry? Why is geometry applicable to the real world?*

Euclid's Five Postulates

1. A straight line can be drawn between any two points.
2. A straight line segment can be extended indefinitely.
3. A circle can be drawn with any center and any radius.
4. All right angles are equal to one another.
5. **The Parallel Postulate:** "If a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side."

? *Notice anything different about the fifth postulate?*

The Parallel Postulate — Visualized

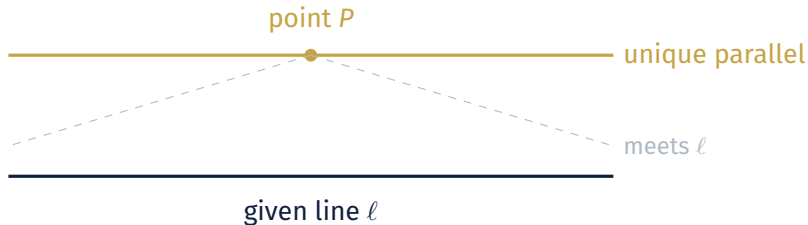


If $\alpha + \beta < 180$, the two lines must eventually meet on this side.

An Equivalent Formulation

Playfair's Axiom (1795)

Given a line and a point not on it, there is **exactly one** line through the point that is parallel to the given line.



Why the Fifth Postulate Is Special

The first four postulates are short, simple, obviously true.

The fifth is:

- **Much more complex**—reads more like a theorem than an axiom.
- **Less self-evident**—claims something about what happens “at infinity.”
- **Not needed for the first 28 propositions**—Euclid avoids it as long as possible.

This led to a question that occupied mathematicians for *two thousand years*:

? *Can the fifth postulate be proved from the other four? Is it really an independent axiom, or a disguised theorem?*

Non-Euclidean Geometry

Two Millennia of Attempts

- Strategy: assume the fifth postulate is *false* and derive a contradiction.
- Every attempt failed—strange consequences appeared, but no outright contradictions.

Saccheri (1733): *Euclid Freed from Every Flaw*

Systematically explored what happens if the fifth postulate is false. Found many bizarre consequences (angle sums $\neq 180$) but no contradiction. Eventually declared the results “repugnant to the nature of a straight line”—hardly a proof!

Alternatives to Postulate 5

Recall Playfair's axiom: through a point not on a line, there is *exactly one* parallel.

5-NONE

Through the point there are **no** parallels.

Leads to **elliptic (spherical)** geometry.

5-MANY

Through the point there are **many** parallels.

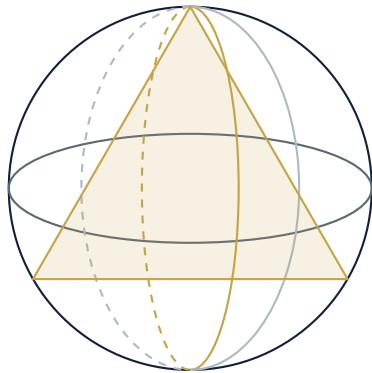
Leads to **hyperbolic (Bolyai–Lobachevsky)** geometry.

The crucial 19th-century discovery: these are not contradictory systems—they are *consistent alternative geometries*.

Geometry of 5-NONE: The Sphere

Think of geometry on the surface of a sphere:

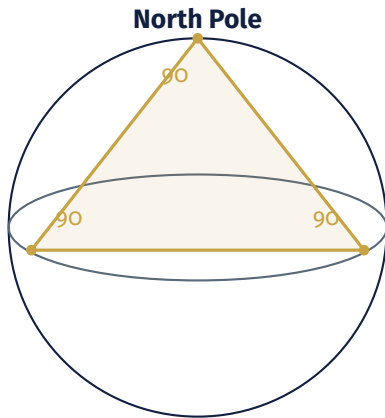
- “Straight lines” = **great circles**
- Any two great circles intersect—**no parallels**
- Triangle angles sum to *more* than 180
- Ratio $c/d < \pi$
- The excess depends on the triangle's *size* relative to the sphere



on a sphere:
angles sum to > 180

Triangle

A Familiar Example: The Earth



Walk south from the North Pole to the equator.

Turn 90 and walk along the equator.

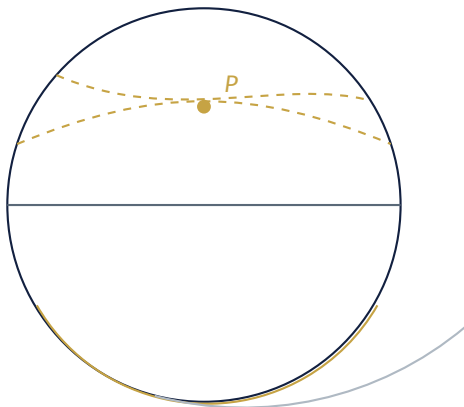
Turn 90 and walk north—back where you started.

Angle sum: $270 > 180$

Geometry of 5-MANY: Hyperbolic Space

Discovered independently by Bolyai (1832) and Lobachevsky (1829):

- **Many** parallels through any point
- Triangle angles sum to < 180
- $c/d > \pi$ (“extra space”)
- No rectangles exist!
- Cannot be perfectly embedded in ordinary 3D space



Poincaré disk model

Multiple “parallels” through P

Comparing the Three Geometries

	Euclidean	Elliptic	Hyperbolic
Parallels through P	exactly 1	0	many
Angle sum of $\triangle$	$= 180$	> 180	< 180
c/d for a circle	$= \pi$	$< \pi$	$> \pi$
Curvature	0 (flat)	positive	negative



flat



positive curvature



negative curvature

Common assumption: constant curvature. Riemann's generalization: curvature can **vary from point to point**.

The Key Discovery

Relative Consistency Proof

If Euclidean geometry is consistent, *then* so are the non-Euclidean geometries. The fifth postulate is **genuinely independent** of the other four.

This was revolutionary:

- There is not *one* possible geometry but *many*.
- Euclid's geometry is not the only *logically possible* description of space.
- Which geometry *actually* describes physical space is an *empirical* question—not settled by pure reason.

The Epistemological Crisis

The existence of non-Euclidean geometries created a profound philosophical problem:

- **Before:** Geometry was the paradigm of *a priori* knowledge—known with certainty by pure reason.
- **After:** There are multiple *consistent* geometries. Pure reason cannot tell us which is true of physical space.

Three philosophical responses:

1. **Kantian:** Euclidean geometry is a necessary precondition for spatial experience.
2. **Empiricist:** The geometry of space is discovered by measurement.
(Helmholtz, Riemann)
3. **Conventionalist:** The choice of geometry involves a conventional element.
(Poincaré)

We will examine these positions next week.

Riemann's Generalization

- All three constant-curvature geometries share an assumption: curvature is the *same everywhere*.
 - Riemann (1854) dropped this assumption: curvature can **vary from point to point**—like the surface of a potato.
 - This requires a local **metric** that specifies distances at each point.
- ? *Sound familiar? The “metric as extra structure” that resolved Zeno’s paradox of infinite divisibility is exactly what Riemann introduced as the foundation of geometry.*
- Riemann’s framework will become essential when we reach Einstein’s general theory of relativity.

For Next Time

- **Read:** Norton, “Philosophy of Space and Time” (online as C 2)—sections on geometry and conventionalism.
- **Think about:**
 - › Can you think of an experiment that would *definitively* determine the geometry of space?
 - › What would Kant say about non-Euclidean geometry?

Next Tuesday: **Conceptions of Geometry**

- Poincaré’s conventionalism and the heated disk
- Reichenbach and coordinative definitions
- The empirical vs. conventional status of geometry

Questions?

chris.smeenk@gmail.com